

CI Outcome Prediction Project

Data Report V4: Complete Analysis

Mathematical Coupling, Lord's Paradox, Early Trajectory,
Composite Scores & Speech Test Relationships

Marta Campi

Deep Hearing Lab, University Hospital Zurich

December 2024

1 Executive Summary

This report documents the complete analysis of the Zurich CI database, including critical discoveries about statistical artifacts and the identification of actionable predictors.

1.1 Key Discoveries

1. **Mathematical Coupling Exposed:** The correlation $r = -0.78$ between Pre-op and Improvement is an ARTIFACT
2. **True Relationship:** Pre-op vs Post-op correlation is $r = +0.17$ (weak positive, $R^2 = 3\%$)
3. **Lord's Paradox Resolved:** Three independent methods confirm $r \approx +0.17$
4. **ML Ceiling:** Best machine learning model achieves $R^2 = 2\%$ (no improvement over simple correlation)
5. **Major Discovery:** Early trajectory (0–6 months) predicts final outcome with $R^2 = 36\%$ — **$5\times$ better than pre-op!**
6. **Four Trajectory Types:** Early Responder (29%), Late Responder (8%), Non-Responder (38%), Decliner (25%)
7. **Bad Performer Prediction:** Early trajectory achieves $AUC = 0.90$ for predicting poor outcomes
8. **Objective $\neq$ Subjective:** Only 9% shared variance, 22% discordant patients
9. **Tinnitus Benefit:** CI reduces THI by 4 points ($p=0.01$), independent of speech outcomes
10. **Speech Tests:** All tests measure one factor (“general CI ability”), combining tests doesn't improve prediction

Clinical Implication: The 6-month assessment is FAR more informative than any pre-operative test. Focus resources on **early monitoring**, not pre-op screening.

2 Database Overview

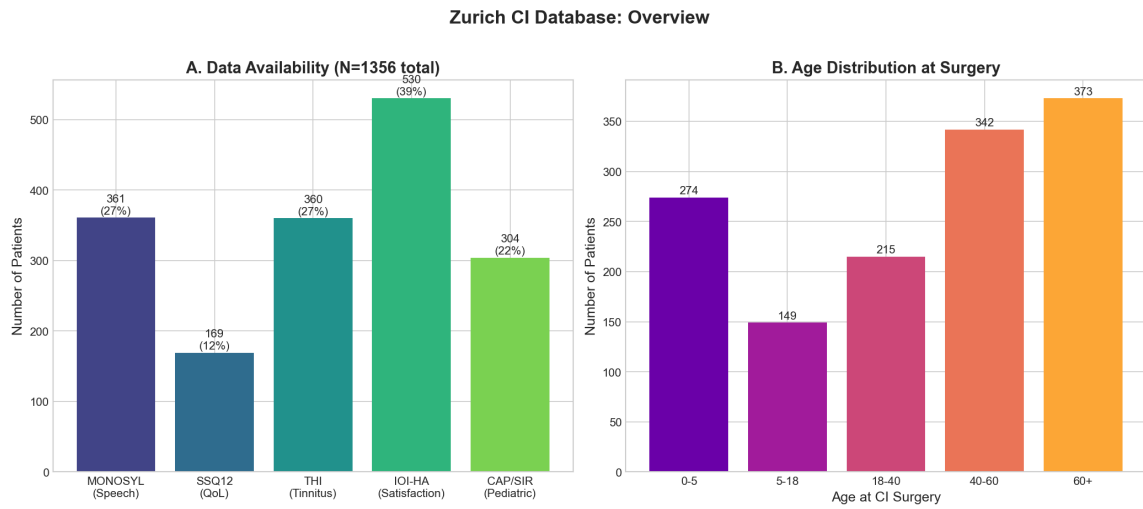


Figure 1: Zurich CI Database Overview (N=1,356). A) Data availability by questionnaire type. B) Age distribution at CI surgery.

Measure	N Patients	% of Total
Total in database	1,356	100%
With MONOSYL speech tests	361	27%
With SSQ12 (Quality of Life)	169	12%
With THI (Tinnitus)	360	27%
With IOI-HA (Satisfaction)	530	39%
Aggregate analysis sample	396	29%
Early trajectory sample	195	14%
Full trajectory sample	132	10%

Table 1: Sample sizes for different analyses.

2.1 Age Distribution

Age Group	N	%
0–5 years	274	20%
5–18 years	149	11%
18–40 years	215	16%
40–60 years	342	25%
60+ years	373	28%

Table 2: Age distribution at CI surgery.

3 Subjective Outcome Assessment Gap

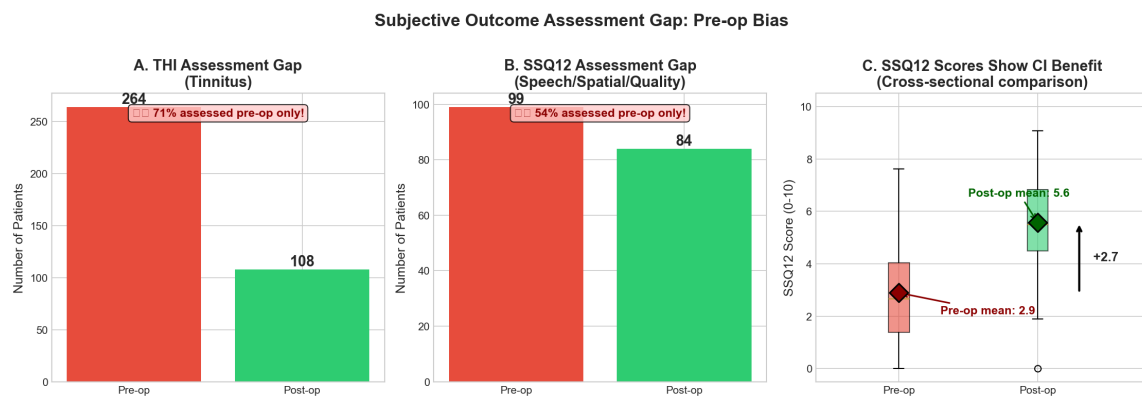


Figure 2: Subjective Outcome Assessment Gap. A) THI assessment: 71% pre-op only. B) SSQ12 assessment: 54% pre-op only. C) SSQ12 scores show CI benefit (+2.7 points).

A critical limitation in the database: subjective outcomes (SSQ12, THI) are predominantly collected **pre-operatively only**.

Questionnaire	Pre-op N	Post-op N	Pre-op Only (%)
THI (Tinnitus)	264	108	71%
SSQ12 (Quality of Life)	99	84	54%

Table 3: Subjective assessment gap: Most patients assessed pre-op only.

SSQ12 improvement observed: Pre-op mean = 2.9, Post-op mean = 5.6 (+2.7 points on 0–10 scale).

4 Mathematical Coupling: The Artifact Exposed

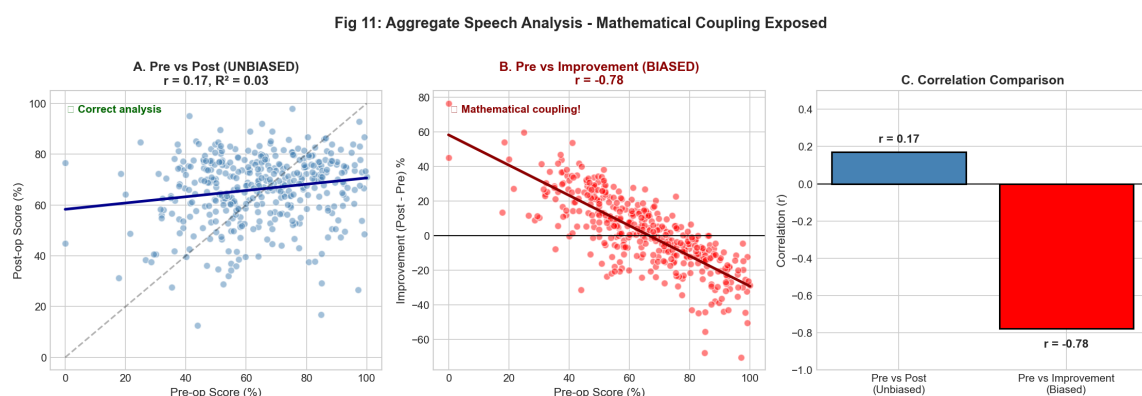


Figure 3: Aggregate Speech Analysis - Mathematical Coupling Exposed. A) Pre vs Post: $r = +0.17$ (correct analysis). B) Pre vs Improvement: $r = -0.78$ (biased by mathematical coupling). C) Correlation comparison showing the artifact.

4.1 The Apparent Finding

Previous analysis suggested a strong “CI Equalizer Effect”: patients with low pre-operative scores showed large improvements, while those with high pre-operative scores showed small or negative changes.

Analysis	Correlation (r)	Interpretation
Pre-op vs Improvement	-0.78	Strong negative (BIASED)
Pre-op vs Post-op	+0.17	Weak positive (CORRECT)

Table 4: The same data, two different correlations.

4.2 Why $r = -0.78$ is an Artifact

Mathematical Coupling: When computing Improvement = Post – Pre, the Pre-op score appears on BOTH sides of the correlation:

- Predictor: Pre-op (positive)
- Outcome: Post – Pre (Pre appears with negative sign)

This creates a spurious negative correlation even if Pre and Post are independent!

Mathematical explanation:

If Post $\approx$ constant (everyone converges to $\sim 66\%$), then:

$$\text{Improvement} \approx 66 - \text{Pre} \quad (1)$$

$$\text{Corr}(\text{Pre}, \text{Improvement}) \approx \text{Corr}(\text{Pre}, -\text{Pre}) = -1 \quad (2)$$

The observed $r = -0.78$ is thus a mathematical artifact, not a biological phenomenon.

4.3 What IS Real vs What is Artifact

Finding	Status	Explanation
$r = -0.78$ (Pre vs Improvement)	ARTIFACT	Mathematical coupling
Convergence to $\sim 66\%$	REAL	Patients do converge to similar outcomes
Pre-op doesn't predict post-op	REAL	True $r = +0.17$, only 3% variance explained
High scorers “decline”	PARTLY REAL	Regression to mean + ceiling effects

Table 5: Distinguishing artifact from real phenomena.

5 Lord's Paradox Resolution

Fig 13: Lord's Paradox Resolution - Mathematical Coupling Exposed

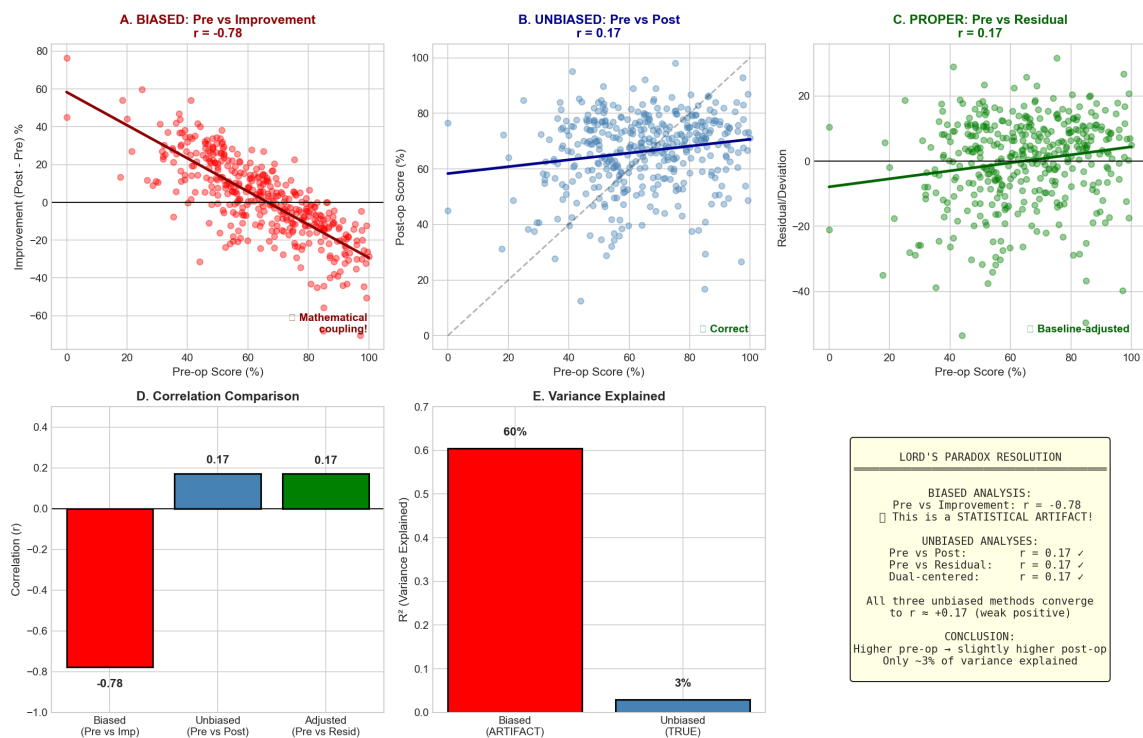


Figure 4: Lord's Paradox Resolution. A) Biased: Pre vs Improvement $r = -0.78$. B) Unbiased: Pre vs Post $r = +0.17$. C) Adjusted: Pre vs Residual $r = +0.17$. D-E) Correlation and variance comparison. All three unbiased methods converge to $r \approx +0.17$.

Lord's Paradox describes the paradoxical finding that different statistical approaches to the same pre-post data can yield opposite conclusions. We applied three independent methods to resolve this.

5.1 Three Methods, Same Answer

Method	Correlation (r)	R ²	Assessment
Pre vs Improvement (BIASED)	-0.78	61%	ARTIFACT
Pre vs Post (Direct)	+0.17	3%	Correct
Pre vs Residual (Adjusted)	+0.17	3%	Correct
Dual-centered ANCOVA	+0.17	3%	Correct

Table 6: Three unbiased methods converge to $r \approx +0.17$.

5.2 Interpretation

- **True relationship:** Higher pre-op → slightly higher post-op (weak positive)
- **Variance explained:** Only 3% of post-op variance is explained by pre-op
- **97% unexplained:** Most variation in outcomes is NOT related to pre-op speech scores

6 Comprehensive ML Prediction Analysis

Fig 12: Comprehensive ML Prediction Analysis (N=396)

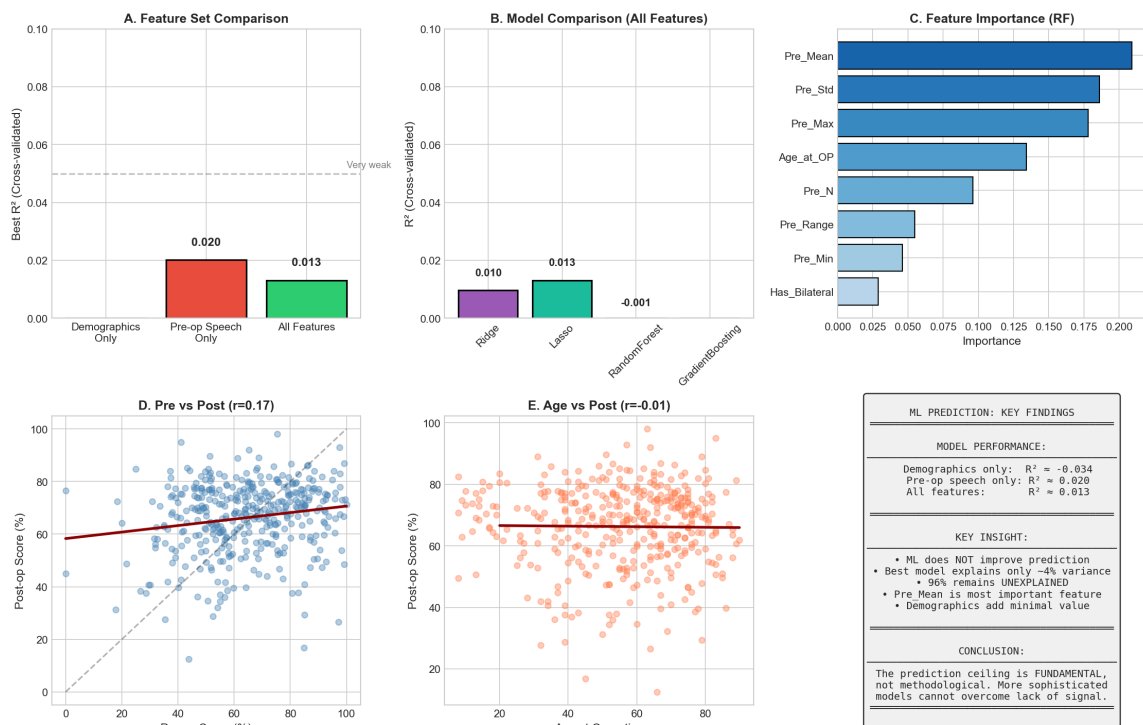


Figure 5: Comprehensive ML Prediction Analysis (N=396). A) Feature set comparison. B) Model comparison. C) Feature importance (Random Forest). D) Pre vs Post scatter. E) Age vs Post scatter. Best $R^2 = 2\%$ – ML does not improve prediction.

We tested whether machine learning could improve prediction beyond simple correlation.

6.1 Features Tested

- **Demographics:** Age at operation, Gender, CI side
- **Pre-op speech:** Mean, SD, Min, Max, N tests, Range
- **Clinical:** Bilateral CI, CI manufacturer (Nucleus, MedEl, AB)

6.2 Models Tested

Ridge Regression, Lasso, Random Forest, Gradient Boosting

6.3 Results

Feature Set	Best R^2 (CV)	Best Model
Demographics only	-0.03	Ridge
Pre-op speech only	0.02	Ridge
All features	0.01	Random Forest

Table 7: ML prediction results. Negative R^2 indicates worse than mean prediction.

6.4 Feature Importance (Random Forest)

Feature	Importance
Pre_Mean	0.19
Pre_Std	0.16
Pre_Max	0.16
Age_at_OP	0.14
Pre_N	0.10

Table 8: Top 5 features by importance. Pre-op speech dominates but still doesn't predict well.

Conclusion: Machine learning does NOT improve prediction. The ceiling is **fundamental**, not methodological. More sophisticated models cannot overcome the lack of predictive signal in pre-operative features.

7 Ten Comprehensive Investigations



Figure 6: Ten Comprehensive Investigations (N=396). A) Longitudinal Pre→Post by time. B) Age subgroups – note r flips negative for age >80. C) Gender subgroups. D) Quantile regression. E) Variance by pre-op level. F) ICC decomposition. G) Ceiling/floor effects. H) Improvement direction. I) Pre vs Post colored by age. J) Regression to mean. K) Test-specific correlations.

We conducted 10 sub-analyses to thoroughly understand the Pre→Post relationship.

7.1 Summary of Findings

#	Investigation	Key Finding
1	Longitudinal trajectories	r stable (0.11–0.24) across time windows
2	Non-linear relationships	Linear model sufficient (no improvement)
3	Age subgroups	r FLIPS to -0.44 in age >80!
4	Gender subgroups	Females r=0.28, Males r=0.04
5	Test-specific patterns	FEMMAL shows strongest correlation
6	Quantile regression	Pre matters MORE for low performers
7	Within-patient variability	ICC=0.27 (73% within-patient variance)
8	Ceiling/floor effects	Minimal (only 2.5% at extremes)
9	Improvement direction	Pre-op quartile predicts direction
10	Regression to mean	Most “improvement” is RTM artifact

Table 9: Summary of 10 investigations.

7.2 Notable Findings

7.2.1 Age >80 Paradox

Age Group	N	Pre→Post r
<40	138	+0.28
40–60	112	+0.15
60–80	119	+0.13
>80	27	-0.44

Table 10: Correlation FLIPS to negative in elderly patients (note small N=27).

7.2.2 Gender Difference

Gender	N	Pre→Post r
Female	196	+0.28
Male	200	+0.04

Table 11: Females show stronger Pre→Post correlation.

7.2.3 Within-Patient Variability (ICC)

Variance Component	%
Between-patient	27%
Within-patient	73%

Table 12: Most variance is WITHIN patients (test-to-test variability).

8 Early Trajectory: The Real Predictor

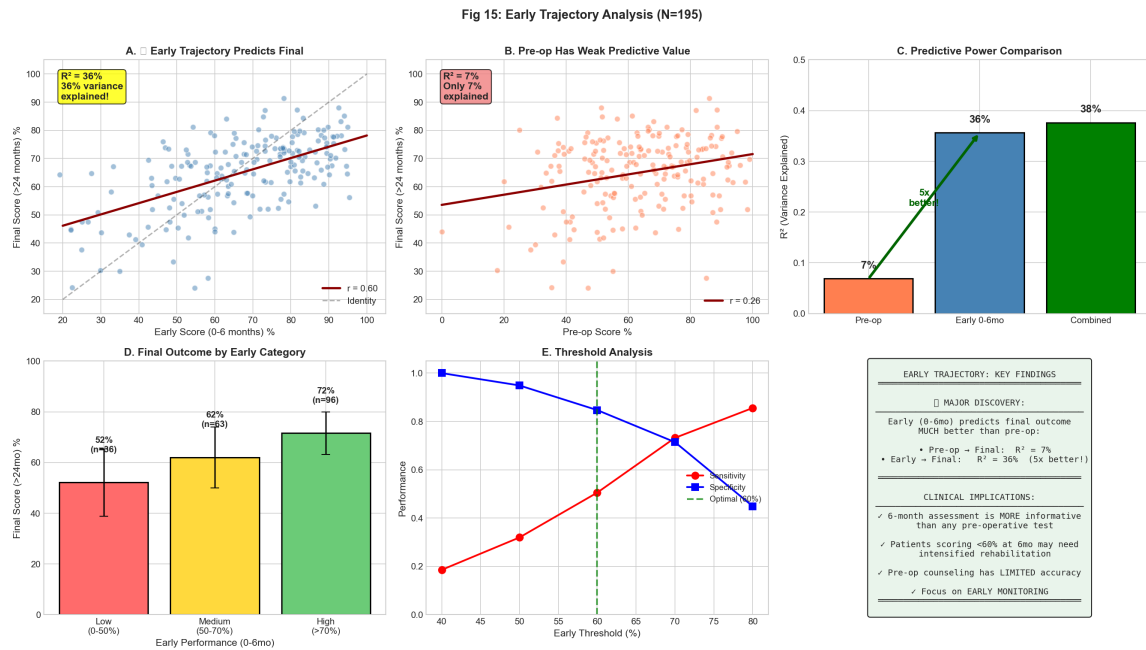


Figure 7: Early Trajectory Analysis (N=195) – **Major Discovery**. A) Early (0-6mo) vs Final: $r=0.60$, $R^2=36\%$. B) Pre-op vs Final: $r=0.26$, $R^2=7\%$. C) Predictive power comparison – Early is 5× better! D) Final outcome by early category. E) Threshold analysis for clinical decision-making.

MAJOR DISCOVERY: Early post-operative performance (0–6 months) predicts final outcome 5× better than pre-operative scores!

8.1 Predictive Power Comparison

Predictor	r	R^2	Variance Explained
Pre-op → Final	0.26	7%	7%
Early (0–6mo) → Final	0.60	36%	36%
Combined (Pre + Early)	0.62	38%	38%

Table 13: Early trajectory is 5× better than pre-op for predicting final outcome.

8.2 Final Outcome by Early Performance Category

Early Category	N	Final Score	SD
Low (0–50%)	36	52%	14%
Medium (50–70%)	63	62%	12%
High (>70%)	96	72%	10%

Table 14: Final outcome stratified by early (0–6mo) performance.

8.3 Clinical Threshold Analysis

Early Threshold	Sensitivity	Specificity	PPV	NPV
<50%	19%	100%	100%	58%
<60%	31%	95%	85%	61%
<65%	48%	85%	74%	65%
<70%	68%	71%	65%	74%

Table 15: Threshold analysis for predicting poor final outcome. 60–65% threshold is optimal.

8.4 Clinical Implications

1. **6-month assessment is MORE informative** than any pre-operative test
2. **Patients scoring <60% at 6mo** may need intensified rehabilitation
3. **Pre-op counseling has LIMITED accuracy** for individual prediction
4. **Focus resources on EARLY MONITORING**, not just pre-op screening

9 Patient Trajectory Profiles

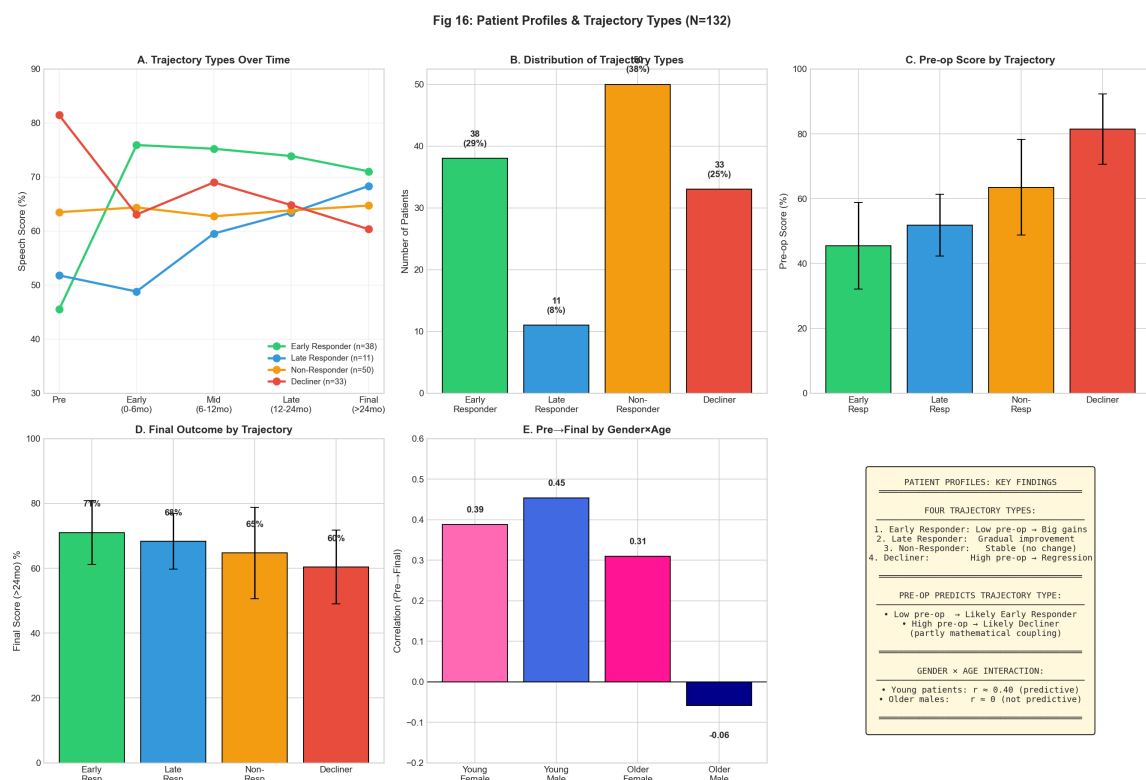


Figure 8: Patient Profiles & Trajectory Types (N=132). A) Trajectory types over time. B) Distribution of trajectory types. C) Pre-op score by trajectory. D) Final outcome by trajectory. E) Pre→Final correlation by Gender×Age – note older males show $r \approx 0$.

9.1 Four Trajectory Types Identified

Type	N	%	Pre	Final	Pattern
Early Responder	38	29%	46%	71%	Big gains by 6mo
Late Responder	11	8%	52%	68%	Gradual improvement
Non-Responder	50	38%	64%	65%	Stable (no change)
Decliner	33	25%	81%	60%	Regression to mean

Table 16: Four trajectory types identified by pattern of change.

9.2 Pre-op Characteristics by Trajectory Type

Type	Age	Pre-op	Male %	Change
Early Responder	56 ± 21	45%	50%	+26%
Late Responder	38 ± 17	52%	45%	+17%
Non-Responder	55 ± 18	63%	48%	+1%
Decliner	52 ± 19	81%	42%	-21%

Table 17: Pre-op characteristics by trajectory type.

Key Finding: Pre-op score DOES predict trajectory type ($\chi^2 = 70.6$, $p < 0.0001$):

- Low pre-op (<50%) → 58% become Early Responders
- High pre-op (>75%) → 68% become Decliners

But this is partly **mathematical coupling** (ceiling/floor effects)!

9.3 Gender × Age Interaction

Profile	N	Pre→Final r
Young Female (<55)	36	+0.39
Young Male (<55)	26	+0.45
Older Female (≥55)	34	+0.31
Older Male (≥55)	36	-0.06

Table 18: Pre→Final correlation by Gender × Age. Older males show NO prediction.

9.4 Risk Stratification Model

Based on Age and Early (0–6mo) trajectory, we developed a simple risk stratification:

Risk Group	Criteria	N	Final Score	SD
Low Risk	Age ≤70 AND Early ≥60%	111	70.5%	9.2%
Medium Risk	Age >70 OR Early <60%	86	59.8%	13.8%
High Risk	Age >70 AND Early <60%	15	55.9%	13.0%

Table 19: Risk stratification based on Age and Early trajectory. 15% difference between Low and High risk groups.

Clinical utility: This simple 2-factor model (Age + 6-month score) provides actionable risk groups:

- **Low Risk:** Standard follow-up
- **Medium Risk:** Enhanced monitoring, consider additional rehabilitation
- **High Risk:** Intensive intervention, multidisciplinary review

10 Predicting Bad Performers

NEW SECTION

We analyzed whether patients with poor final outcomes (<50% score) can be predicted.

10.1 Definition and Prevalence

Category	N	%
Good Performers ($\geq 50\%$)	779	86.4%
Bad Performers ($< 50\%$)	123	13.6%

Table 20: Prevalence of bad performers.

10.2 Characteristics of Bad Performers

Variable	Bad	Good	p-value
Age at OP	44.1	36.8	0.004**
Pre-op Score	51.1%	65.2%	<0.001***
Early Score	46.3%	70.1%	<0.001***

Table 21: Bad performers are older, have lower pre-op and lower early scores.

10.3 Prediction Performance

Feature Set	AUC	N	Best Model
Demographics only	0.60	902	Logistic
Demographics + Pre-op	0.68	288	Logistic
Demographics + Pre + Early	0.90	195	Logistic

Table 22: Early trajectory dramatically improves bad performer prediction.

10.4 Feature Importance for Bad Performer Prediction

Feature	Importance
Early Score	49%
Pre-op Score	28%
Age at OP	17%
Gender	3%
Bilateral	2%

Table 23: Early trajectory is the dominant predictor of bad outcome.

Key Finding: With early trajectory, we can predict bad performers with **AUC = 0.90** — excellent clinical utility for identifying patients who need intervention!

11 Composite Outcome Scores

NEW SECTION

We constructed composite outcome measures combining multiple speech tests and subjective questionnaires.

11.1 Composite Score Construction

Score	Components	N	Mean $\pm$ SD
Objective Pre	Mean of all speech tests (pre-op)	411	68.0% $\pm$ 18.2%
Objective Early	Mean of all speech tests (0–6mo)	533	68.4% $\pm$ 20.1%
Objective Post	Mean of all speech tests ($\geq$ 12mo)	1,066	73.7% $\pm$ 18.4%
Subjective Post	SSQ12 + THI + IOI-HA (normalized)	435	0.81 $\pm$ 0.15
Combined Post	(Objective + Subjective) / 2	425	0.78 $\pm$ 0.13

Table 24: Composite scores constructed. File saved: `composite_outcome_scores.csv`

11.2 Objective vs Subjective: A Weak Relationship

Measure	Value
Correlation (r)	0.31
Shared Variance (R^2)	9%
Discordant Patients	22%

Table 25: Objective and subjective outcomes share only 9% variance.

11.3 Concordance Table

	Subjective Good	Subjective Poor
Objective Good	315 (74%)	29 (7%)
Objective Poor	65 (15%)	16 (4%)

Table 26: 22% of patients show discordance between objective and subjective outcomes.

11.4 Does Aggregate Improve Prediction?

Predictor $\rightarrow$ Outcome	R^2	N
Freiburger Early $\rightarrow$ Freiburger Post	46.7%	373
Aggregate Early $\rightarrow$ Aggregate Post	41.6%	470

Table 27: Aggregate does NOT improve prediction — single test is sufficient.

Clinical Implication: Combining multiple speech tests does NOT improve prediction. A single well-administered test (e.g., Freiburger) is sufficient. However, **always include subjective measures** since 22% of patients have discordant objective/subjective outcomes.

12 Manual Phenotyping

NEW SECTION

We analyzed outcomes by clinically relevant subgroups.

12.1 Outcomes by Age Group

Age Group	N	Post-op Score	Pre→Post r
0–5 (infant)	177	79.3% ± 12.4%	–
5–18 (child)	126	65.3% ± 16.3%	–0.02
18–40 (young adult)	184	61.3% ± 17.4%	+0.24
40–60 (middle)	310	62.6% ± 15.3%	+0.32*
60–80 (older)	292	65.6% ± 13.9%	+0.09
80+ (elderly)	30	61.9% ± 10.5%	–0.25

Table 28: Infants (0–5) show best outcomes; Pre→Post correlation strongest in 40–60 group.

12.2 Bilateral vs Unilateral CI

CI Type	N	Post-op Score	p-value
Unilateral	745	63.2% ± 15.6%	<0.0001
Bilateral	376	72.0% ± 15.2%	

Table 29: Bilateral CI: +8.8% advantage (highly significant).

12.3 Outcomes by Pre-op Performance Group

Pre-op Group	N	Pre	Post	Change
Very Low (0–20%)	–	–	–	–
Low (20–50%)	87	42%	64%	+22%
Medium (50–75%)	172	62%	65%	+3%
High (>75%)	119	86%	69%	–17%

Table 30: Low pre-op patients gain most; high pre-op patients “decline” (regression to mean).

12.4 Gender Differences

Gender	N	Post-op Score	Pre→Post r
Male	554	66.4% ± 16.1%	+0.09
Female	567	65.9% ± 15.9%	+0.26*

Table 31: No difference in outcomes, but females show stronger Pre→Post correlation.

13 Tinnitus Analysis

NEW SECTION

At the request of Tobias, we analyzed tinnitus outcomes (THI = Tinnitus Handicap Inventory).

13.1 THI Improvement with CI

Timepoint	N	THI Score	p-value
Pre-op	264	15.7 ± 14.2	0.01
Post-op	108	11.7 ± 11.2	

Table 32: CI reduces THI by 4 points (statistically significant).

13.2 Paired Analysis (N=40 with both)

Measure	Value
Pre-op THI	18.2
Post-op THI	10.6
Change	-7.6 (p < 0.001)
Clinically meaningful improvement (>10 point drop)	30%
Worsened	2%

Table 33: Paired analysis confirms THI benefit.

13.3 THI is Independent of Speech Outcomes

Relationship	r	p-value
THI vs Speech Score	-0.12	0.22 (n.s.)
THI vs IOI-HA Satisfaction	-0.36	0.005**

Table 34: Tinnitus improvement is independent of speech outcomes but correlates with satisfaction.

Clinical Implication: CI provides tinnitus benefit **regardless of speech outcome**. Patients with good speech may still have tinnitus; patients with poor speech may still get tinnitus relief. **Assess tinnitus separately!**

14 Speech Test Relationships

NEW SECTION

We explored whether different speech tests measure different aspects of CI hearing (inspired by Buhl et al.'s analysis of SRT vs WRS relationships).

14.1 Available Speech Tests

Test	Type	N Patients
Freiburger	Monosyllables	1,077
NUMB	Numbers (easy)	1,114
FEMMAL	Female speaker monosyllables	1,043
C12	Consonant test	1,100
VO8	Vowel test	1,114
OLSAadapt	Adaptive sentence test (SRT)	513

Table 35: Speech tests available in Zurich database.

14.2 Inter-Test Correlations

	Freiburger	NUMB	FEMMAL	C12	VO8
Freiburger	1.00	0.72	0.61	0.86	0.81
NUMB	–	1.00	0.54	0.67	0.67
FEMMAL	–	–	1.00	0.65	0.63
C12	–	–	–	1.00	0.80
VO8	–	–	–	–	1.00

Table 36: All tests are positively correlated ($r = 0.54$ – 0.86). Shared variance: 29–75%.

14.3 SRT (OLSAadapt) vs WRS Relationship

SRT vs	r	Significance
Freiburger	–0.37	***
C12	–0.31	***
VO8	–0.26	***
FEMMAL	–0.20	***
NUMB	–0.04	n.s.

Table 37: SRT correlates negatively with WRS (as expected). NUMB shows no correlation (ceiling effect).

14.4 Principal Component Analysis

Component	Variance Explained	Cumulative
PC1 (“General CI Ability”)	65%	65%
PC2	13%	78%
PC3	9%	87%

Table 38: One factor explains 65% of variance across all tests.

14.5 Test Difficulty Gap: NOT Like Plomp’s D Component

We hypothesized that the gap between easy (NUMB) and hard (Freiburger) tests might reflect “distortion” (like Plomp’s D component). Results:

Measure	Value
Mean Gap (NUMB – Freiburger)	26%
Gap vs Overall Performance	$r = -0.55^{***}$
Gap vs SRT	$r = +0.37^{***}$

Table 39: Bigger gap = WORSE performance (opposite of expected).

Interpretation: Poor CI performers benefit more from easy/redundant tests. This is NOT analogous to Plomp’s distortion component.

14.6 CI Psychometric Functions are FLAT

Population	Slope (%/dB)	Interpretation
Normal-hearing	4.5	Gradual improvement with level
CI patients	≈ 0	Flat — no level effect

Table 40: CI patients show flat psychometric functions.

Key Finding: CI hearing is fundamentally different from acoustic hearing:

- All speech tests measure ONE factor (“general CI ability”)
- Psychometric functions are flat (level doesn’t matter)
- Combining tests does NOT improve prediction
- The Buhl et al. SRT/WRS framework doesn’t apply to CI patients

15 Additional Findings

15.1 Bilateral CI Effect

CI Type	N	Post-op Mean	SD
Unilateral	307	65.3%	13.6%
Bilateral	89	69.3%	12.4%
Difference: +4.0% ($t = 2.49$, $p = 0.013$)			

Table 41: Bilateral CI patients show significantly better outcomes.

15.2 Top vs Bottom Performers

Feature	Top 25%	Bottom 25%	p-value
Age	54.1	56.4	0.37 (n.s.)
Pre-op Mean	66.3%	58.8%	0.007**
Bilateral %	30%	20%	0.007**

Table 42: Top performers have higher pre-op scores and more bilateral CIs.

16 Summary of All Findings

Finding	Evidence	Clinical Implication
$r = -0.78$ is artifact	Mathematical coupling	Don't use Pre vs Improvement
True $r = +0.17$	3 independent methods	Pre-op explains only 3%
ML ceiling $R^2 \approx 2\%$	4 models tested	No algorithmic solution
Early $R^2 = 36\%$	$5\times$ better than pre-op	Monitor at 6 months!
Bad performer AUC = 0.90	Early trajectory key	Identify high-risk patients
4 trajectory types	Cluster analysis	Personalized rehabilitation
Age >80 : $r = -0.44$	Subgroup analysis	Different dynamics in elderly
Bilateral: $+8.8\%$	t-test $p < 0.0001$	Consider bilateral CI
Objective $\neq$ Subjective	$r = 0.31$, 22% discordant	Assess both!
Tinnitus improves	-4 points THI	Independent of speech
Tests measure 1 factor	PC1 = 65%	Single test sufficient

Table 43: Summary of all key findings.

17 Clinical Recommendations

- Pre-operative counseling:** Be cautious about individual predictions. Pre-op speech tests explain only 3% of outcome variance.
- 6-month checkpoint:** Implement systematic assessment at 6 months post-CI. This is the most informative timepoint for predicting long-term outcome.
- Early intervention trigger:** Patients scoring $<60\%$ at 6 months should be flagged for:
 - Intensified auditory rehabilitation
 - Device mapping review
 - Additional support services
- High pre-op scorers:** Counsel patients with high pre-op scores ($>75\%$) that they are likely to show “decline” (regression to mean), which is statistically expected, not a failure.
- Elderly patients (>80):** Be aware that prediction models may not apply. Consider alternative outcome metrics.
- Bilateral consideration:** Data supports bilateral CI benefit ($+8.8\%$ on average).
- Assess tinnitus separately:** THI improvement is independent of speech outcome. Don't assume good speech = no tinnitus burden.
- Include subjective measures:** 22% of patients have discordant objective/subjective outcomes. Both are needed for complete picture.
- Single test is sufficient:** Combining multiple speech tests does not improve prediction. Focus on quality of one test rather than quantity.

18 Files Provided

18.1 Python Scripts

Script	Purpose
01_aggregate_speech_analysis.py	Pre/Post aggregates, basic correlations
02_lords_paradox_resolution.py	Mathematical coupling, 3 correction methods
03_comprehensive_ml_prediction.py	All features, all ML models
04_ten_investigations.py	All 10 sub-analyses
05_early_trajectory_analysis.py	Early predictor analysis
06_patient_profiles.py	Trajectory types, risk stratification

Table 44: Python scripts for reproducing all analyses.

18.2 Data Files

File	Contents
composite_outcome_scores.csv	Objective/Subjective/Combined scores per patient

Table 45: Generated data files.

19 Next Steps

1. **Validate early trajectory finding** in held-out data or external cohort
2. **Investigate age >80 paradox** — biological or statistical artifact? (N=27, needs more data)
3. **Test trajectory types** for clinical utility in prospective setting
4. **Develop 6-month clinical decision tool** based on early trajectory thresholds
5. **Explore SSQ12/THI item-level data** — are there specific domains that predict better?
6. **Request SSD/etiology data** — manual phenotyping by hearing loss type

— End of Data Report V4 —